Gaia DR3 Validation Request

The single most important missing piece

This file contains the formal ADQL query needed to validate the main result of the paper.

If you can execute this query, please open an Issue or Pull Request with your results.

Query

```
SELECT COUNT(*) AS n_gk_stars,
       AVG(distance_gspphot) AS mean_dist_pc,
       AVG(teff_gspphot) AS mean_teff
FROM gaiadr3.gaia_source
WHERE distance_gspphot BETWEEN 138 AND 168
  AND l BETWEEN 76 AND 92
  AND b BETWEEN -6 AND 10
  AND teff_gspphot BETWEEN 3700 AND 6000
```

Extended query (individual stars)

```
SELECT TOP 200
    source_id, ra, dec, l, b,
    distance_gspphot,
    teff_gspphot,
    logg_gspphot,
    mh_gspphot,
    phot_g_mean_mag,
    parallax,
    parallax_over_error
FROM gaiadr3.gaia_source
WHERE distance_gspphot BETWEEN 138 AND 168
  AND l BETWEEN 76 AND 92
  AND b BETWEEN -6 AND 10
  AND teff_gspphot BETWEEN 3700 AND 6000
  AND parallax_over_error > 5
ORDER BY distance_gspphot ASC
```

Archive access

- ESA Gaia Archive: https://gea.esac.esa.int/tap-server/tap
- Web interface: https://gea.esac.esa.int/archive/

Cross-match (step 2)

After obtaining the source_id list from Gaia DR3, cross-match with:
- NASA Exoplanet Archive: https://exoplanetarchive.ipac.caltech.edu
- TESS Input Catalog: https://tess.mit.edu/science/tess-input-catalogue/

What we are looking for

Stars of type G (Teff 5200-6000 K) or K (Teff 3700-5200 K) at approximately 450-550 light-years (138-168 parsecs) within 8° of galactic coordinates l=84°, b=2° (equatorial: RA=310.1°, Dec=+45.1°).

Interpretation guide

Result	Meaning
0 G/K stars found	Zone may be obscured by dust — check extinction
60-100 G/K stars, none with planets	Expected baseline — hypothesis remains open
G/K star with confirmed planet in HZ	Very significant — contact author immediately
G/K star with TESS candidate (unconfirmed)	Significant — worth follow-up

Important note

A negative result (no planets found) does NOT falsify the main result of the paper (the geometric match d=0.452, p≈0). It simply means we cannot yet identify a specific host candidate. The geometric result stands independently of what lies at that location.

A positive result would make the paper considerably more interesting.

Please open a GitHub Issue with your query results. Any outcome is welcome.